

1. What is the correct way to declare a constant in ES6?

- a) `var PI = 3.14;`
- b) `let PI = 3.14;`
- c) `const PI = 3.14;`
- d) `PI = 3.14;`

2. Which ES6 feature allows extracting values from arrays or objects into distinct variables?

- a) Arrow Functions
- b) Template Literals
- c) Destructuring
- d) Spread Operator

3. What will be the output of the following code?

```
let a = 10;  
let b = 20;  
[a, b] = [b, a];  
console.log(a, b);
```

- a) 10 20
- b) 20 10
- c) 10 10
- d) 20 20

4. Which of the following is an arrow function in ES6?

- a) `function() { return x; }`
- b) `() => x`
- c) `() -> x`
- d) `function => x`

5. What does the spread operator (...) do in ES6?

- a) Combines two arrays into one
- b) Expands an iterable (array, string) into individual elements
- c) Creates a deep copy of an object
- d) Both a and b

6. Which ES6 feature allows embedding expressions in strings using backticks?

- a) String Interpolation
- b) Template Literals
- c) String Concatenation
- d) Tagged Templates

7. What is the output of the following code?

```
const greet = (name = "Guest") => `Hello, ${name}!`;
console.log(greet());
```

- a) Hello, Guest!
- b) Hello, undefined!
- c) Hello, !
- d) Error

8. Which keyword is used to declare block-scoped variables in ES6?

- a) var
- b) let
- c) const
- d) Both b and c

9. What is a Promise in ES6 used for?

- a) Synchronous operations
- b) Handling asynchronous operations
- c) Looping through arrays
- d) Defining classes

Answer: b) Handling asynchronous operations

Explanation: Promises handle async tasks like API calls with `.then()` and `.catch()`.

10. What will this ES6 class code output?

```
class Animal {  
  constructor(name) {  
    this.name = name;  
  }  
  speak() {  
    return `${this.name} makes a noise.`;  
  }  
}  
const dog = new Animal("Dog");  
console.log(dog.speak());
```

- a) Dog barks.
- b) Dog makes a noise.
- c) undefined makes a noise.
- d) Error

11. What is TypeScript primarily used for?

- a) Adding dynamic typing to JavaScript
- b) Adding static typing to JavaScript
- c) Replacing JavaScript entirely
- d) Making JavaScript run faster

12. Which keyword is used to define a custom type in TypeScript?

- a) interface
- b) type
- c) class
- d) Both a and b

- interface is used for object shapes.

- `type` can define aliases for primitives, unions, tuples, etc.

```
type Age = number;  
interface Person { name: string; age: Age; }
```

13. What is the output of this TypeScript code?

```
let x: number | string = 10;  
x = "20";  
console.log(typeof x);
```

- a) `number`
- b) `string`
- c) `number | string`
- d) Error

14. How do you define an optional property in a TypeScript interface?

- a) `property?: type`
- b) `property: type | undefined`
- c) `property: optional type`
- d) `property: type?`

Example:

```
interface User {  
  name: string;  
  age?: number; // Optional  
}
```

15. What are Generics in TypeScript used for?

- a) To enforce type safety in arrays
- b) To create reusable components that work with multiple types
- c) To replace interfaces
- d) To optimize runtime performance